Macroeconomics II (Spring 2026)

Lecture 4: AD–AS in Equations

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A short dictionary: diagrams $\leftrightarrow$ equations

From Lecture 3 to Lecture 4

- Lecture 3: AD–AS diagrams classify shocks by *comovement* of (Y, P).
- Lecture 4: write a compact set of equations that reproduces the same logic.
- Two objects will carry almost all intuition:
  - **Output gap** $x_t \equiv y_t - y_t^{FE}$ (tight vs slack economy)
  - **Expected price level** p_t^e (what firms/households had in mind when setting wages/prices)

Level vs rate (so we don't confuse P and π later)

- **Price level:** $p_t \equiv \log P_t$.
- **Inflation:** $\pi_t \equiv p_t - p_{t-1}$.
- A one-time increase in P_t is a level effect; persistent inflation means π_t stays high for several periods.
- In this lecture we keep the AD–AS vertical axis as P **(or p)** to match the figures.
- Next lecture we switch to π to study inflation dynamics explicitly.

Output gap and unemployment (connection only)

- We keep output-gap notation throughout: $x_t \equiv y_t - y_t^{FE}$.

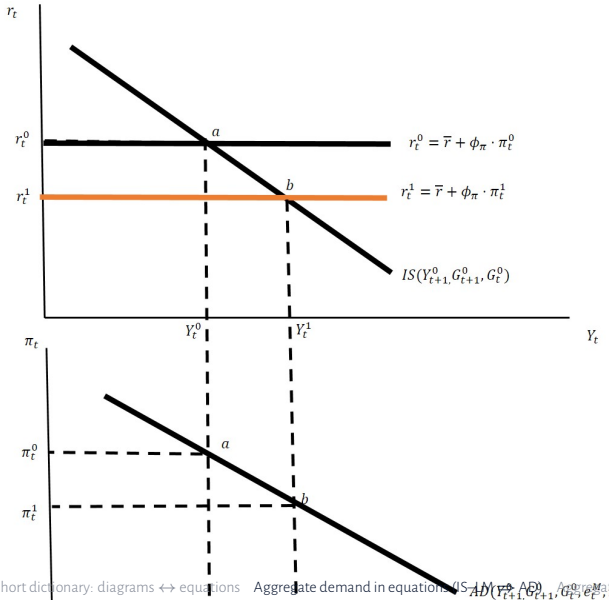
- Labor market tightness is often summarized by the unemployment gap:

$$u_t - u_t^n \approx -\gamma\, x_t \qquad (\gamma > 0) \quad \text{(Okun's law)}.$$

- So:
  - $x_t > 0$ (output above potential) $\Rightarrow$ unemployment below natural
  - $x_t < 0$ (output below potential) $\Rightarrow$ unemployment above natural

Aggregate demand in equations (IS–LM $\Rightarrow$ AD)

AD from IS–LM: the one mechanism



- Fix nominal money supply M.
- Higher P lowers real balances M/P.
- Lower M/P shifts LM left $\Rightarrow$ interest rate rises.
- Higher interest rate reduces spending $\Rightarrow$ lower Y.

A simple IS–LM system (linear)

- **IS (goods market):** output falls when the real rate rises

$$Y = \bar{A} - ar, \qquad a > 0,$$

where $\bar{A}$ summarizes autonomous spending (preferences, G, taxes, etc.).

- **LM (money market):**

$$\frac{M}{P} = kY - hi, \qquad k, h > 0.$$

- **Fisher (in the background):** $i = r + \pi^e$. (For now take π^e as given.)

Solve IS + LM for $Y(P)$ (the AD equation)

From LM: $i = \frac{k}{h}Y - \frac{1}{h}\frac{M}{P}$, so

$$r = i - \pi^e = \frac{k}{h}Y - \frac{1}{h}\frac{M}{P} - \pi^e.$$

Plug into IS: $Y = \bar{A} - ar$,

$$Y = \bar{A} - a\left(\frac{k}{h}Y - \frac{1}{h}\frac{M}{P} - \pi^e\right)$$

$$\left(1 + a\frac{k}{h}\right)Y = \bar{A} + a\pi^e + \frac{a}{h}\frac{M}{P}.$$

- Therefore Y increases in real balances M/P and decreases in P.
- This is the **downward slope of AD** in (P, Y) space.

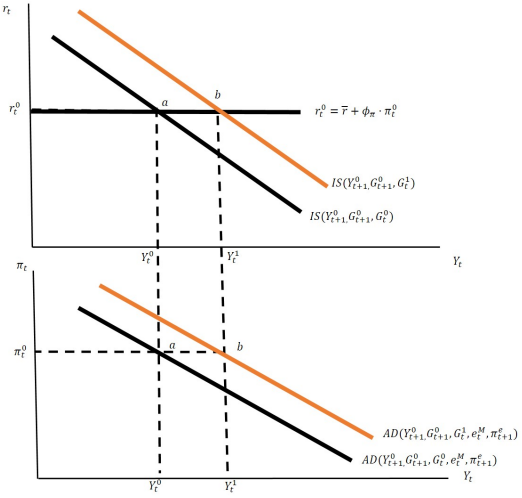
What shifts AD (in equations)

- AD is summarized by

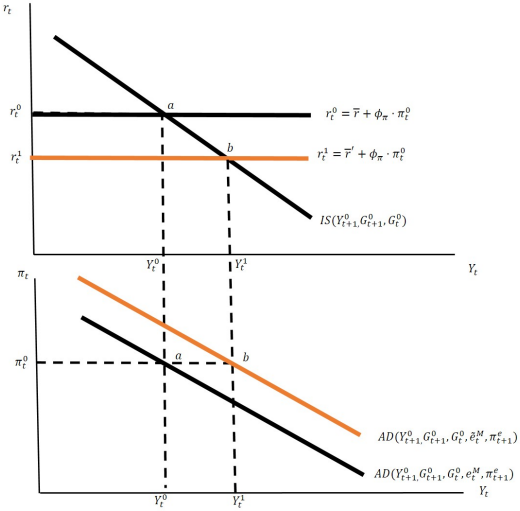
$$Y(P) = \frac{\bar{A} + a\pi^e + \frac{a}{h}\frac{M}{P}}{1 + a\frac{k}{h}}.$$

- **AD shifts right** when, at each P, equilibrium Y rises:
  - $\bar{A} \uparrow$ (e.g., $G \uparrow$, lower taxes, optimism about future income)
  - $M \uparrow$ (higher nominal money supply)
  - $\pi^e \uparrow$ (given i, lower real rate; in the IS–LM derivation it enters through Fisher)

Two textbook shifts (same as Lecture 3)



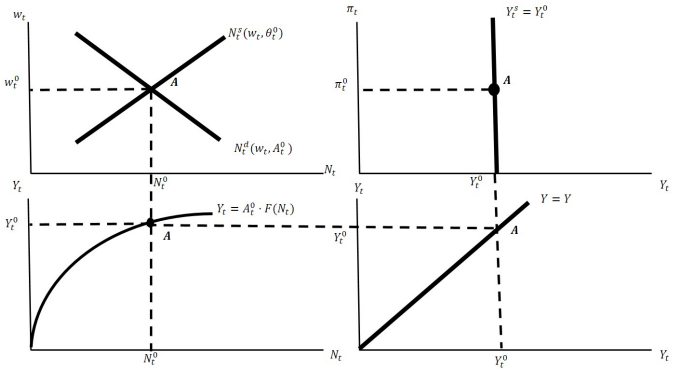
$G \uparrow$ (or $\bar{A} \uparrow$): AD shifts right



$M \uparrow$: real balances higher $\Rightarrow$ AD shifts right

Aggregate supply in equations (SRAS + FE)

FE as the long-run supply benchmark



- FE output Y^{FE} comes from the RBC supply block (labor market + production).
- In the long run, prices adjust so output returns to Y^{FE}.
- FE shifts with technology, factor supply, and capital.

SRAS as "price surprises move output" (level form)

A convenient reduced form (Lucas/expectations-augmented supply):

$$y = y^{FE} + \alpha\,(p - p^e), \qquad \alpha > 0.$$

- If $p > p^e$ (prices higher than expected), real wages are lower than expected $\Rightarrow$ firms hire more $\Rightarrow$ output rises.
- If $p < p^e$, output falls below y^{FE}.
- The slope of SRAS in (p, y) space is $1/\alpha$.

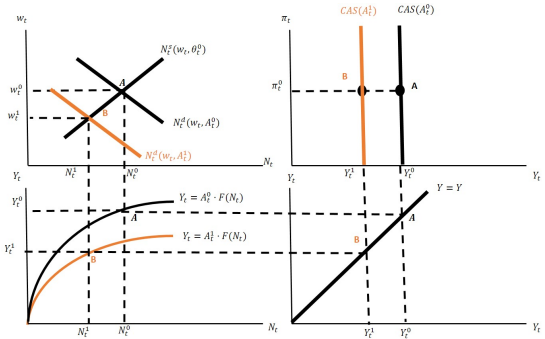
SRAS written in output-gap notation

Define output gap $x \equiv y - y^{FE}$. Then

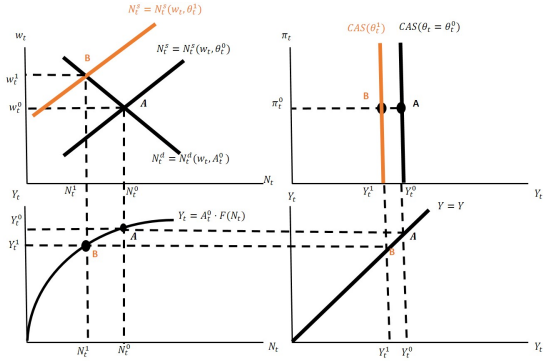
$$x = \alpha\left(p - p^e\right).$$

- Output gap is proportional to the **price-level surprise** $\left(p - p^e\right)$.
- A left shift of AS in Lecture 3 corresponds to shocks that raise marginal cost for given x, or (equivalently) raise the p needed to produce a given y.
- Next lecture we will rewrite this in inflation form and specify how p^e evolves (expected inflation).

Supply-side shifts in the figures



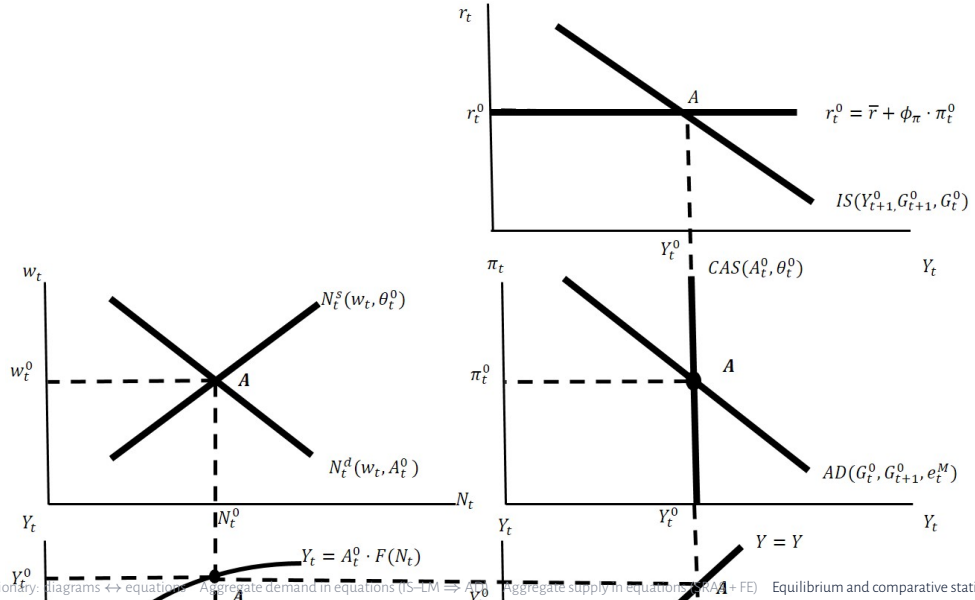
$A \downarrow$ shifts FE / CAS left



labor supply $\downarrow$ shifts FE / CAS left

Equilibrium and comparative statics

AD–AS equilibrium (in one picture)



AD–AS equilibrium (solve once)

Use a simple AD reduced form $y = \bar{y}_D - \phi\, p$ with $\phi > 0$ (downward slope), and SRAS $p = p^e + \lambda(y - y^{FE})$ with $\lambda > 0$ (upward slope).

- Substitute AD into SRAS:

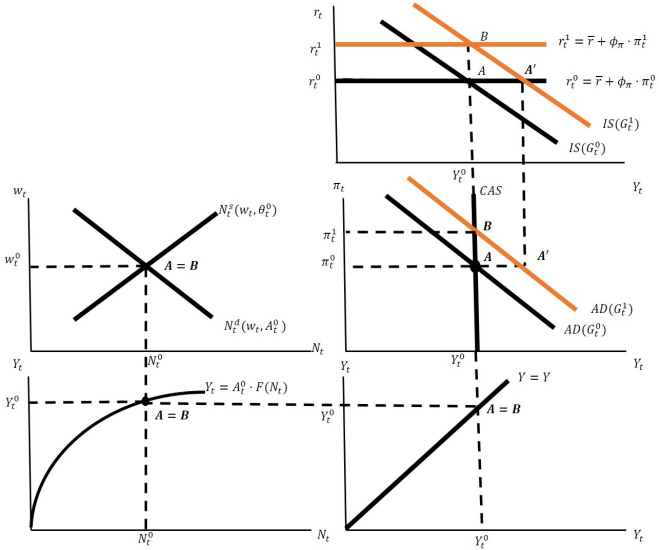
$$p = p^e + \lambda\big(\bar{y}_D - \phi p - y^{FE}\big).$$

- Solve for p:

$$(1 + \lambda\phi)p = p^e + \lambda(\bar{y}_D - y^{FE}) \quad \Rightarrow \quad p = \frac{p^e + \lambda(\bar{y}_D - y^{FE})}{1 + \lambda\phi}.$$

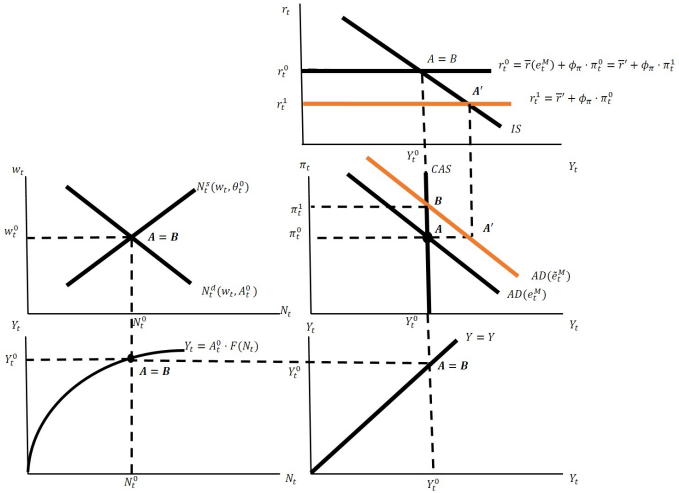
- Then y follows from AD: $y = \bar{y}_D - \phi p$.

Demand expansion: why (Y, P) move together



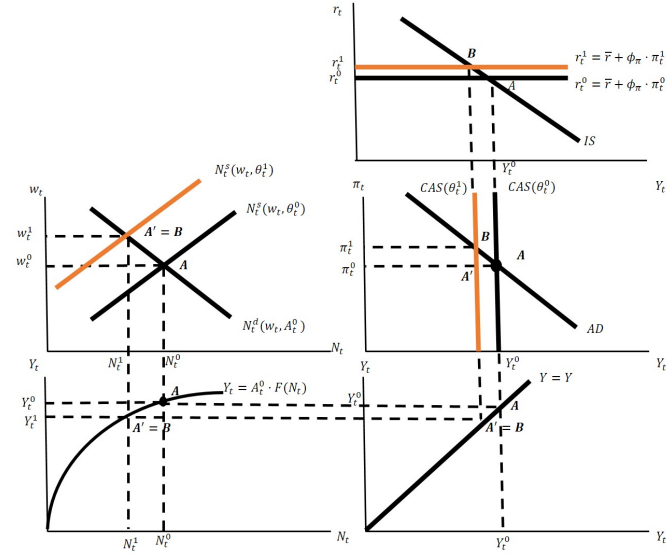
- Demand expansion means $\bar{y}_D \uparrow$ (AD shifts right).
- From the solved expression: p rises when $\bar{y}_D$ rises.
- AD then implies y rises as well.

Monetary easing: one parameter shift, same logic



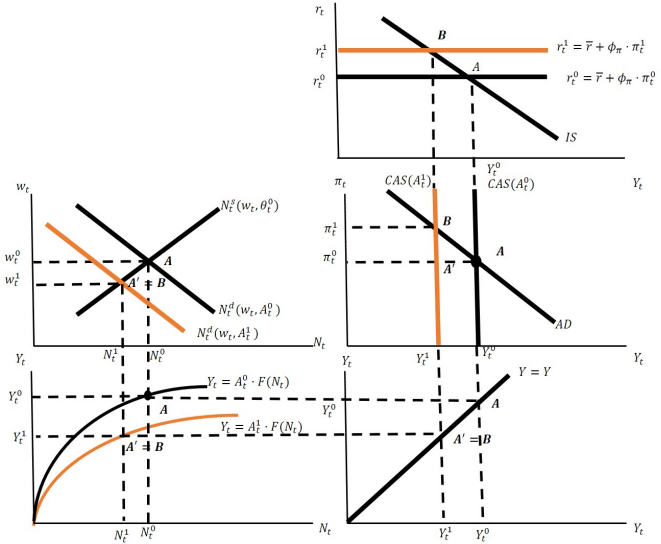
- Monetary easing shifts AD right (for each P, higher Y).
- In the IS–LM derivation, this is $M \uparrow$ or lower i (higher real balances / lower real rate).
- Equilibrium: (Y, P) rise together in the short run.

Supply contraction: stagflation in one line



- Supply contraction means $y^{FE} \downarrow$ or AS shifts up/left.
- In the solved expression: if y^{FE} falls, p rises.
- But lower y^{FE} also pushes output down.

Medium run: expected price level catches up



- SRAS depends on p^e. When p^e updates, SRAS shifts.
- Medium-run logic: $p^e \uparrow$ until output returns to y^{FE}.
- This is the bridge to next lecture: specify how p^e evolves $\Rightarrow$ expected inflation.

One-page summary

Cheat sheet: what each symbol means in the math AD–AS

- **Demand side (AD):** $y = \bar{y}_D - \phi p$ (downward in p)
  - $\bar{y}_D$ collects *demand shifters*: G, taxes, wealth/optimism, monetary stance (M, i), etc.
- **Supply side (SRAS):** $p = p^e + \lambda(y - y^{FE})$ (upward in y)
  - y^{FE} is potential output from the RBC supply block (technology, labor, capital).
  - p^e is the expected price level; updating p^e shifts SRAS over time.
- **Unemployment link:** $u - u^n \approx -\gamma x$ (Okun).

Next lecture (without naming “Old Keynesian” yet)

- Replace p^e with a rule for expected inflation (how expectations are formed).
- Replace the level-form SRAS with an inflation form (Phillips curve language).
- Introduce central bank practice: inflation targeting, Taylor rules, and how a policy rule shifts AD.